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# From pilots to decision systems: embedding generative AI into strategic decision-making through a socio-technical and governance lens

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**ABSTRACT**

Generative AI (GAI) promises superior analytics and agility in strategy work, yet organisations struggle to move beyond pilots towards routinised decision inputs. This study investigates how GAI becomes embedded in strategic decision-making (SDM) through a qualitative single-case analysis of a global multi-brand group, based on 27 semi-structured executive interviews triangulated with internal documents and industry reports. Structured inductive coding yields a process model identifying enablers, leadership-driven adoption, quick wins, prompt-based experimentation, workforce training, secure platforms, and dedicated investments, and barriers such as strategic ambiguity, limited awareness, hallucination risks, prompt-engineering deficiencies, data readiness, and privacy or IP concerns. The analysis specifies a four-stage pathway comprising Awareness and Exploration, Experimentation and Pilots, Formal Adoption and Integration, and Institutionalisation and Transformation, with admission gates for quality, provenance, explainability, and accountability. Human-in-the-loop arrangements redistribute responsibilities between AI and managers, while governance templates and socio-technical alignment determine whether GAI outputs are admitted into formal deliberation. Findings reframe GAI not as an autonomous oracle but as decision support within decision systems, clarifying conditions under which creative or efficient outputs become strategically admissible. The study contributes a socio-technical, governance-anchored model that addresses the pilot-to-decision gap and offers actionable heuristics for scaling GAI responsibly in strategic decision-making.

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Generative AI; strategic decision-making; decision support; socio-technical systems; human-in-the-loop; governance

## 1. Introduction

The rapid development of Generative Artificial Intelligence (GAI) significantly influences strategic management across organisational contexts (Dwivedi et al., 2023; Kshetri et al., 2023). Across functions and industries, GAI provides novel methods for analysing large data volumes, refining competitive strategies, and supporting complex decision-making (Kshetri et al., 2023). Beyond efficiency gains, GAI raises fundamental questions about effectively integrating humagn judgement with AI-driven analyses to achieve innovative strategic solutions (Dwivedi & Wang, 2022; Hossain et al., 2022). Recent advancements, particularly transformer-based models such as OpenAI ChatGPT, facilitate deeper integration of AI into strategic processes by rapidly interpreting data and enhancing decision-making methodologies (Burger et al., 2023). Positioning this study at the intersection of GAI and strategic decision-making (SDM), we examine how GAI interacts with multi-stakeholder decision contexts, governance requirements, and accountability regimes, and how such interactions reshape SDM (Csaszar et al., 2024).

Current literature highlights the transformative potential of GAI in business-model innovation (Kanbach et al., 2024; Schwaeke et al., 2024), talent acquisition (Roppelt et al., 2024; Roppelt, Schuster, et al., 2025), and healthcare (Roppelt, Jenkins, et al., 2025), particularly through generating data-driven insights and improving predictive decisions (Molina et al., 2024). Prior work largely focuses on consumer-facing use cases (Huh et al., 2023); by contrast, we know less about how GAI is integrated into organisational strategic decision-

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making with multiple stakeholders and longer decision cycles (Athaide et al., 2025). Such contexts often feature layered decision structures, heterogeneous data landscapes, and interdependent roles. Despite initial enthusiasm, many firms still struggle with moving beyond isolated GAI use-cases due to data privacy, legal ambiguities, and ethical governance challenges (Bag et al., 2021; Rana et al., 2024). Additionally, uncertainty remains regarding how to meaningfully incorporate GAI into SDM requiring long-term planning, accountability, and context-specific expertise (Susarla et al., 2023). AI-based approaches, such as those used for strategic forecasting, make an important contribution to reducing uncertainty in the decision-making process (Acuña et al., 2025). At the same time, the wide range of applications for GAI, such as in marketing, healthcare, and manufacturing, illustrates its disruptive impact across industries (Ooi et al., 2025). Together, these strands suggest substantial potential, yet we still lack integrative explanations of how this potential becomes routinised in formal strategic choices under real organisational constraints. Despite technical advances, we still lack a comprehensive explanation of how GAI outputs become admissible evidence in SDM under real-world governance and capability constraints.

GAI refers to model-based systems, typically large-scale transformer architectures, that autonomously generate novel outputs such as text, images, or code, thereby supporting analytical, creative, and communicative tasks in organisations (Kshetri et al., 2023; Yehia, 2025). SDM denotes multi-layered decision processes oriented towards long-term organisational goals, where alternatives are generated, evaluated, and committed to under uncertainty, often with high relevance for resource allocation and competitive positioning (Atuahene-Gima & Li, 2004; Payne & Frow, 2014). Integration describes the systematic embedding of GAI into organisational routines, infrastructures, and governance structures, ensuring that pilots evolve into scalable, reliable capabilities, consistent with socio-technical perspectives emphasising the co-evolution of technology, processes, culture, and competences (Sony & Naik, 2020; Vial, 2019). Human – AI Interaction encompasses collaborative arrangements, such as human-in-the-loop systems, in which explainability, transparency, and trust frame the use of AI-generated outputs while ultimate accountability remains with human decision-makers (Glikson & Woolley, 2020; Jarrahi, 2018; Miller, 2019; Shrestha et al., 2019). In the following, these constructs are used consistently, with GAI understood as an augmentation of human judgement in SDM and integration conceptualised as a socio-technical and governance-bound process.

Beyond technical feasibility, scholars emphasise understanding how GAI interacts with organisational routines, culture, and human agency (Dwivedi & Wang, 2022; Jarrahi, 2018; Shrestha et al., 2019). Attention focuses particularly on mechanisms enabling or hindering adoption, the role of trust and explainability (Ehsan et al., 2021; Glikson & Woolley, 2020), and how GAI reshapes cognitive roles between humans and AI. Gaps persist regarding how firms operationalise GAI in distributed, functionally diverse environments, and how this affects strategic responsibilities. Two perspectives dominate the current debate yet remain insufficiently reconciled: on the one hand, research and managerial discourse portraying GAI as a lever for superior strategic insight and agility; on the other, mounting evidence of organisational frictions such as governance constraints, liability concerns, data fragmentation, and capability asymmetries that prevent scaling beyond pilots. This unresolved tension constitutes the central gap: organisations are urged to rely on GAI to improve strategic choices, while the socio-organisational conditions of SDM systematically impede such usage.

To further illustrate this gap, we contrast recurrent claims in the GAI and decision-systems discourse with the organisational realities observed in our case. The mapping, detailed in Section 2 (Table 1), shows how widely cited promises confront governance, provenance, data, and capability frictions in practice, and how our framework advances the debate.

Building on this mapping, we now specify our research questions, directly targeting the identified admission bottlenecks and governance frictions. The study's main goal is to explain how generative AI becomes a routine input for strategic decision-making in complex, multi-stakeholder organisations. To achieve this, we will specify the integration process, the structures that enable or hinder this process, and the human-AI interaction dynamics that influence strategic choices:

> (1) How is GAI incorporated into SDM processes within complex organizational settings? (2) Which structures and mechanisms enable or hinder the integration of GAI into strategic decision-making and adjacent management processes within multi-brand or divisional environments? (3) How do human actors and GAI interact in shaping strategic choices and organizational transformation in decentralized ecosystems?

**Table 1.** Recurrent claims in the GAI and management literature versus organisational realities in decision systems.

| No. | What the literature often claims? | Where the tension arises in practice? | What our data show? | Implication and contribution of this paper |
|---|---|---|---|---|
| 1 | GAI boosts content performance and efficiency across the commercial funnel (e.g. messaging, assets) (Hartmann et al., 2025) | Strategic admissibility of outputs remains unclear under governance and reputational thresholds | Pilots improve content creation, but outputs are blocked before entering strategic forums | Process model specifying gates that move GAI outputs from experimentation to admissible strategic inputs |
| 2 | AI adoption is accelerating with broad value pools (McClure et al., 2024) | Adoption concentrates in analytics, not in deliberative SDM | Tool use is wide but shallow; hand-offs to decision forums fail on explainability | Clarifies the alignment problem between analytical affordances and SDM governance |
| 3 | AI foundations emphasize role reconfiguration and scaling (McClure et al., 2024) | Role change is contested, especially for mid-level managers as translators of AI insights | Emerging "insight owner" roles lack clarity and accountability | Specifies boundary conditions for accountable human-in-the-loop roles |
| 4 | Digital practices reshape organizational processes (Schmitt et al., 2021) | Local improvisations conflict with central governance for data and compliance control | Local teams use GAI ad hoc; central units block diffusion | Explains how governance checkpoints shape routinization paths |
| 5 | AI creates "superhuman" outputs (Hartmann et al., 2025) | Creative or efficient does not equal strategic admissibility | Creative outputs rejected without traceability and compliance | Separates creative efficacy from strategic admissibility, offering evaluative criteria |
| 6 | AI will transform decision-making (Davenport et al., 2020; Kumar, Ashraf, et al., 2024) | Integration stalls at the interface of provenance, explainability, and accountability | Decision episodes halt when provenance or responsibility is ambiguous | Specifies heuristics for admitting GAI outputs into formal strategic deliberation |

Consistent with theory-building case research, we develop boundary-aware propositions rather than universal claims.

To address this tension empirically, we examine LuxCorp (anonymised), a global multi-brand group with decentralised decision structures. The setting typifies multistakeholder decision networks where local autonomy and central oversight coexist, providing a critical testbed for the routinisation of GAI in strategic work. The case offers access to governance mechanisms, data compliance practices, and cross-functional coordination. Empirical data were assembled from multiple sources to ensure triangulation and validity. We conducted 27 semi structured interviews with senior executives, functional leaders in marketing, technology experts, and legal professionals directly involved in GAI initiatives. Interviews were complemented with internal documents and external industry reports on AI adoption to reconstruct decision episodes and trace integration paths. For analysis, we applied a structured inductive coding approach comprising first-order informant terms, second-order themes, and aggregate dimensions, following established practices for theory building in qualitative research (Gioia et al., 2013). The coding linked emergent categories to the theorised tension and yielded a grounded framework that identifies organisational enablers and barriers, characterises human AI interaction patterns, and outlines stages of operationalisation of GAI within SDM. Our inductive stance combines data-driven theorising with sensitising concepts from socio-technical integration and trust in AI.

This study contributes theoretically by clarifying how organisational structures, collaborative dynamics, and governance practices shape GAI integration in strategic decision-making. First, we explicate the alignment problem between GAI's analytical affordances and distributed SDM governance, thereby resolving the identified tension and extending socio-technical views of digital transformation to GAI-enabled strategy work. Second, we articulate a process model of GAI operationalisation in SDM that traces progression from pilots to scaled capabilities, specifying role changes between human actors and AI across decision phases. Third, we delineate trust- and explainability-related mechanisms (e.g. boundary conditions for human-in-the-loop (HITL) accountability) that condition whether GAI outputs are admitted into formal strategic deliberation. Practically, the paper offers managers and policymakers a structured framework for embedding GAI into strategic workflows. For practitioners, our findings detail capability prerequisites, governance checkpoints, and integration heuristics to move from isolated use-cases to reproducible strategic impact across organisational settings. Overall, while GAI enables agile, data-driven, anticipatory decision-making, its successful integration depends on deliberate organisational alignment across governance, capabilities, and culture, thereby converting experimentation into sustained strategic advantage rather than sporadic efficiency gains.

## 2. Conceptual background

In a decision systems perspective, strategic decision making refers to multi-level, governance bound processes in which problems are framed, alternatives are generated and evaluated, and commitments are made under uncertainty (Jarrahi, 2018; Russell et al., 2020; Seeber et al., 2020; Ulrich et al., 2025). Decision systems integrate analytical artefacts with human oversight, apply transparent criteria for admissibility and responsibility, and target sustained value creation evidenced by measurable outcomes and risk aware allocation (Csaszar et al., 2024; Heilig & Scheer, 2023; Jarrahi, 2018; Seeber et al., 2020; Shrestha et al., 2019; Vial, 2019). Within this overarching decision-systems architecture, we treat socio-technical systems (STS), HITL, and explainable AI (XAI) as complementary layers rather than competing theories: STS provides the contextual co-evolution of technology and organisation, HITL allocates accountable oversight, and XAI supplies epistemic transparency for admissibility (Buijsman, 2024; Goktas, 2024).

To align with the study's mechanism-seeking and gate-oriented lens on SDM, we selected our theoretical triad: STS, HITL, and XAI. Previous organisational studies have shown that socio-technical integration, accountability-preserving human oversight, and the quality of explanations jointly predict whether AI outputs will be used in formal decision-making processes (Ehsan et al., 2021; Jarrahi, 2018; Miller, 2019; Shrestha et al., 2019; Vial, 2019). These lenses correspond to the system, role, and artefact layers, directly addressing observed governance and capability frictions (Bag et al., 2021; Rana et al., 2024). However, adoption-focused, capability-based, and field-level lenses, such as Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT) (Davis & Venkatesh, 1996), dynamic capabilities (Teece et al., 1997), and institutional theory (DiMaggio & Powell, 1983), are less useful because they emphasise individual intention, firm-level renewal, and legitimacy dynamics rather than micro-level admissibility gates. These gates, such as provenance, explainability, and accountability, determine whether AI outputs are used in decision-making processes (Ehsan et al., 2021; Miller, 2019; Ribeiro et al., 2016; Seeber et al., 2020).

Despite the growing adoption of GAI, companies often lack standardised governance, robust data quality regimes, and methods to control for biases and 'hallucinations' (Kshetri, 2024; Yaprak, 2024). Organisational variables such as leadership, risk appetite, and cross-functional collaboration remain critical (Geels, 2004; Sony & Naik, 2020). Recent work on digital transformation similarly shows that dynamic capabilities, digital leadership, and a trust-based innovation culture are decisive for embedding advanced technologies into organisational structures (Bouncken & Kraus, 2022; Held et al., 2025). To analyse these issues and structure-specific business challenges, we use STS (Sony & Naik, 2020) to explore how human-technology interfaces shape GAI adoption. This perspective clarifies why purely technical GAI implementations typically underperform without corresponding socio-cultural alignment. In the following, STS, HITL, and XAI are treated as interlocking components of the decision-systems architecture: STS explains socio-organisational alignment, HITL anchors accountability, and XAI ensures transparency and evidentiary robustness.

### *2.1. Hybrid human – AI decision model*

We define GAI as large-scale transformer-based models that autonomously generate text, images, or code (Kshetri et al., 2023; Yehia, 2025) and conceptualise it as an augmentative capability embedded in socio-technical decision systems under XAI and HITL governance. In organisational settings, AI rarely replaces human judgement; rather, it enhances it. The resulting hybrid model combines AI's analytical capabilities with human creativity, contextual understanding and ethics. STS emphasises the importance of striking a balance between people, technology, culture and processes (Sony & Naik, 2020). Explainable AI (XAI) makes the reasoning behind models transparent, thereby meeting the trust requirements of hybrid decision systems (Wohleber et al., 2021). Trust, transparency and accountability are prerequisites, and XAI, audits and clear governance reduce the friction associated with adoption (Haupt et al., 2025; Kartikeya, 2022; Miller, 2019; Schmidt et al., 2020; Wohleber et al., 2021). Continuous prompt logging ('AI snapshots'), role-specific training and human review loops operationalise this trust at scale (Burger et al., 2023).

Recent evidence highlights that acceptance of GAI in decision support depends not only on transparency but also on perceived benefits and privacy risks. Huynh (2024) shows that functional, social, emotional and epistemic values increase perceived benefits of disclosure, while information sensitivity raises perceived risks and information control reduces them. Benefits strengthen trust, risks undermine it, and trust in turn predicts

users' appreciation of GAI. This mechanism clarifies under which conditions hybrid collaboration is trusted and complements governance-based trust building.

Deepening collaboration also reshapes roles. For example, marketers shift towards strategic orchestration and analysts towards AI interpretation, while decision makers must integrate AI insights with ethical and strategic considerations (Goudira, 2024; Patil, 2024; Schwaeke et al., 2024). Such mechanisms illustrate how perceived value and risk shape trust in hybrid human – AI systems, directly linking to our admission gates of provenance, explainability and accountability. Operationally, HITL instantiates governance (accountable oversight and sign-off), XAI instantiates explainability (traceable, confidence-calibrated rationales), and STS situates both within organisational routines; together they realise the admission gates of quality, provenance, explainability, and accountability inside the decision-systems architecture.

### 2.2. Strategic decision-making

Strategic decision-making (SDM) in organisations involves high complexity, formal approval processes, and multiple professional stakeholders. It combines iterative, data-driven analysis with scenario planning and strategic-fit checks, while engaging both internal and external stakeholders to improve alignment and risk management (Atuahene-Gima & Li, 2004; Payne & Frow, 2014; Reichstein, 2019). Recent literature underlines that SDM is not a linear sequence of steps but a multifaceted system that spans structured and unstructured activities and integrates organisational learning, strategic discourse, and board-level oversight into a coherent architecture across different organisational contexts (Ateljević & Alfirević, 2025).

From a decision-systems perspective, SDM can be understood as a socio-technical arrangement in which data, models, and human judgement are coupled within auditable workflows under explicit governance. Human-in-the-loop configurations ensure that ultimate responsibility remains with human actors, while transparency and explainability determine whether system outputs are strategically admissible (Jarrahi, 2018; Miller, 2019; Shrestha et al., 2019). Provenance, explainability, and accountability thus operate as admission gates that govern the inclusion of AI-generated outputs in formal deliberation, rather than technical accuracy or efficiency alone (Buijsman, 2024; Goktas, 2024; Zeiser, 2024). This interpretation resonates with established research on trust, human oversight, and the organisational embedding of intelligent systems (Bevilacqua et al., 2025; Heilig & Scheer, 2023; Jarrahi, 2018; Ribeiro et al., 2016; Seeber et al., 2020; Shrestha et al., 2019). In our framing, these gates function as ex ante design parameters of decision episodes, derived from HITL governance (accountability) and XAI standards (explainability and provenance) within the decision-systems scaffold.

Recent empirical work reinforces these dynamics. Huynh (2024) shows that organisational users assess GAI outputs not only by their technical accuracy but also through socio-cognitive filters such as trust, explainability, and perceived appreciation. Trust mediates whether GAI recommendations are actually considered in decision settings, while contextual risk and governance safeguards shape admissibility thresholds. These findings substantiate our conceptualisation of admission gates: quality, provenance, explainability, accountability, as prerequisites for GAI outputs to progress from pilots to formally acknowledged decision inputs. To concretise how these gates bind organisational practice, Table 1 contrasts recurrent claims in the GAI discourse with decision-system realities observed in our case and shows how our framework advances the debate.

SDM also constitutes a discursive and social practice. Deliberative arenas structure, authorise, and record strategic actions, thereby shaping how power, expertise, and meaning flow into decisions. Boards of directors and supervisory bodies can enhance decision quality by demanding alternative scenarios and ensuring escalation mechanisms for high-stakes choices. Governance and structure therefore act as quality filters that determine which analytical artefacts, including outputs from AI systems, are admitted into decision forums (Hendry, 2000; Jezak, 2010).

A wide range of tools and techniques support SDM by addressing both technical and social requirements. Multi-Criteria Decision Analysis (MCDA) and scenario analysis help surface trade-offs, uncertainty, and stakeholder preferences, but require adaptation when used at the strategic level, particularly in terms of facilitation and value modelling (Montibeller & Franco, 2010). Strategic analysis instruments connect external and internal perspectives and render assumptions auditable, linking directly to admission gates that condition the entry of AI outputs into decision episodes (Ahmed et al., 2014; Papulova & Gazova, 2016). These

insights align with recent syntheses that emphasise the need for hybrid approaches combining analytical rigour with discursive validation in order to scale SDM across diverse organisational settings (Ateljević & Alfirević, 2025).

Organisational culture and structure remain critical. Innovation-oriented, transparent, and trust-building climates accelerate both the quality of SDM and the assimilation of AI (Kumar et al., 2025). When viewed through the socio-technical systems lens, the scaling of GAI requires that technical readiness and cultural readiness evolve in tandem. This conceptual position clarifies the research gap and motivates our empirical inquiry into how GAI becomes embedded in organisational decision systems (Heilig & Scheer, 2023; Seeber et al., 2020; Shrestha et al., 2019; Vial, 2019).

Despite advances in decision systems, we lack empirically grounded accounts of how GAI outputs become admissible evidence within organisational deliberation. Existing studies emphasise technical performance, yet the decisive filters are provenance, explainability, and accountability under human in the loop governance. We do not know how these gate conditions are operationalised across decision episodes with multiple stakeholders and formal approval routines. Our study addresses this by tracing the socio-technical embedding of GAI and specifying the gate logic that moves outputs from pilots to formally acknowledged decision inputs.

### 2.3. Synthesising STS, HITL, and XAI

We integrate STS, HITL, and XAI into a single, decision-systems-based lens through which to view admissibility. STS provides system fit through routines, governance, and infrastructures, and it explains how organisations embed AI under governance and capability frictions (Heilig & Scheer, 2023; Seeber et al., 2020; Sony & Naik, 2020; Vial, 2019). HITL allocates decision-making accountability and escalation procedures in high-stakes settings. Empirical studies demonstrate that oversight structures influence the adoption of intelligent system outputs (Heilig & Scheer, 2023; Jarrahi, 2018; Seeber et al., 2020; Shrestha et al., 2019). XAI provides understandable explanations and provenance at the time of decision-making. Studies have linked the type of explanation and social transparency to acceptance and use in decision-making processes (Ehsan et al., 2021; Miller, 2019; Ribeiro et al., 2016).

The system, role, and artefact layers are complementary yet also surface practical tensions, such as speed versus auditability (Heilig & Scheer, 2023; Miller, 2019), local autonomy versus accountability (Seeber et al., 2020), and usefulness of explanations versus cognitive load (Ehsan et al., 2021; Ribeiro et al., 2016; Shrestha et al., 2019). We used this synthesis to guide our analysis. Admissions occur when system fit, accountable ownership, and usable explanation/provenance co-occur. Stalls, on the other hand, indicate a deficiency in at least one dimension. This is consistent with documented governance and capability frictions (Bag et al., 2021; Rana et al., 2024). We use STS, HITL, and XAI as sensitising concepts that operationalise this integrated lens in concrete decision-making situations.

## 3. Methodology

Given the exploratory aim of this study, we adopted a qualitative single-case research design (Yin, 2018) to investigate how GAI is integrated into SDM processes. This approach is particularly suited for studying complex, evolving phenomena within real-work organisational environments, where the boundaries between context and phenomena are often fluid (Yin, 2018). To ensure methodological rigour and enable theory development grounded in empirical insight, we applied a systematic inductive approach that facilitates the identification of conceptual patterns through iterative coding, structuring, and interpretation of qualitative data (Gioia et al., 2013).

### 3.1. Data sampling

We purposefully selected LuxCorp (anonymised), a leading global luxury group operating under a decentralised, multi-brand governance structure. While the corporate holding provides strategic, legal, and technological infrastructure, each brand maintains significant autonomy over operational and marketing decisions. This hybrid structure, combining centralised oversight with decentralised innovation, offered an

ideal setting to study the integration of GAI into SDM, particularly regarding the coordination mechanisms, organisational adaption, and digital transformation pathways.

To ensure relevance and richness of insight, we applied purposeful sampling targeting individuals directly involved in strategic planning, digital innovation, and AI governance. Our sample included senior executives (e.g. CTO, CIO), department directors (e.g. IT security, HR), managers, and specialised professionals (e.g. Legal Counsel, Data Analytics Head) from both the holding company and eight independent brands. These brands represented diverse functions such as marketing, product management, IT, operations, and Research & Development (R&D).

This sampling strategy enabled a multi-perspective view of GAI adoption, capturing variation across hierarchical roles, digital maturity levels, and organisational contexts. The divers sample supports an in-depth understanding of how GAI integration unfolds across different strategic domains and socio-technical settings. An overview of interviewees and key characteristics is provided in Table 2.

### 3.2. Data collection

To investigate how GAI is integrated into SDM, how organisational structures facilitate or constrain this integration, and how human actors interact with GAI systems, we collected in-depth qualitative data from multiple sources. The primary dataset comprises 27 semi-structured expert interviews, conducted between February and April 2025, with decision-makers and strategic professionals across multiple business units within LuxCorp.

Participants were selected through purposeful sampling based on their direct involvement in digital transformation programs, GAI-related initiatives, or governance roles. This includes senior leaders, functional managers, and specialists from domains such as IT, legal, human resources, product development, and marketing. Interviews were conducted via video conferencing to ensure flexibility and accessibility across global locations. The average interview length was 25 minutes.

To ensure consistency and theoretical depth, we used a semi-structured interview guidelines informed by preliminary literature on AI adoption, human-AI collaboration, and decision systems. The interview protocol covered the themes including strategic relevance of GAI in the organisation, enablers and barriers of GAI integration, human-AI interaction patterns and trust mechanisms, and organisational governance structures

**Table 2.** Overview of interviewees and characteristics – source: authors' own work.

| Interviewee | Position | Gender | Department | Brand/Unit | Duration of the Interview |
|---|---|---|---|---|---|
| R1 | Manager | m | Marketing | Holding | 25 |
| R2 | Chief of Staff | m | Strategy | Holding | 26 |
| R3 | Manager | m | R&D | Holding | 39 |
| R4 | Chief Information Security Officer | m | IT Security | Holding | 31 |
| R5 | Head | f | R&D | Holding | 43 |
| R6 | Chief Legal Counsel | m | Legal | Holding | 20 |
| R7 | Head | m | Data & Analytics | Holding | 19 |
| R8 | Senior Legal Counsel | m | Legal | Holding | 15 |
| R9 | Group Director | f | HR | Holding | 25 |
| R10 | Chief Technology Officer | m | IT | Holding | 25 |
| R11 | Director | m | Strategy | Holding | 30 |
| R12 | Group Director | m | Marketing | Holding | 33 |
| D1 | Head | m | R&D | AccessoriesA | 30 |
| M1 | Manager | m | Marketing | AccessoriesB | 25 |
| M2 | Director | m | IT | AccessoriesB | 19 |
| C1 | Manager | f | Product Management | JewelryA | 18 |
| C2 | Senior Manager | f | Marketing | JewelryA | 20 |
| A1 | Head | m | Marketing | WatchA | 25 |
| A2 | Head | m | IT | WatchA | 21 |
| I1 | Manager | m | IT | WatchB | 20 |
| I2 | Associate Director | m | Marketing | WatchB | 20 |
| I3 | Associate Director | m | Operations | WatchB | 20 |
| I4 | Global HR Director | m | HR | WatchB | 23 |
| J1 | Manager | m | R&D | WatchC | 13 |
| V1 | Manager | m | R&D | WatchD | 31 |
| W1 | Chief Information Officer | m | IT | WatchE | 25 |
| W2 | Chief Product Officer | m | Product Management | WatchE | 23 |

and ethical safeguards. An overview of these themes is provided in Table 3. The full interview protocol is available in the Appendix (see Table A1).

To enable triangulation and enrich contextual understanding, we supplemented interview data with internal and external sources. These included internal documents (e.g. AI governance frameworks, GAI policy presentations, training materials), official company reports, and publicly available material such as press releases and industry analyses. This multi-modal data collection strategy enhanced the reliability of our findings and allowed for a richer interpretation of GAI integration as a complex, context-sensitive phenomenon. An overview of the collected data is provided in Table 4.

### 3.3. Data processing

All data, encompassing both interviews and supplementary materials, were analysed using the qualitative data analysis software MAXQDA24. All interviews audio-recorded and transcribed using MacWhisper. Transcripts were reviewed manually to ensure accuracy and completeness. Participants anonymity was preserved by assigning identifiers (e.g. Interviewee C1, Interviewee R1), and any explicit company reference were anonymised.

For qualitative data analysis, we followed an inductive and systematic approach well-suited for generating grounded theoretical insights from rich qualitative data (Gioia et al., 2013). In alignment with this approach, open coding was first conducted on the interview transcripts to identify informant-centric concepts (1st-order concepts). These were clustered during the axial coding phase into more abstract researcher-centric categories (2nd -order themes), which in turn were synthesised into aggregated dimensions (design dimensions) capturing the core aspects of GAI integration into strategic decision-making processes.

To ensure analytical rigour and trustworthiness, the coding was performed collaboratively by all authors. Each author independently coded initial subset of transcripts to identify preliminary themes. The emerging codebooks were then compared and discussed in joint coding sessions during which conceptual overlaps and divergens were resolved through iterative dialog and negotiated agreement, rather than relying on statistical inter-coder reliability measures. This interpretive approach follows established practice in inductive qualitative research, where conceptual convergence is prioritised over quantitative agreement (Gioia et al., 2013; Pratt et al., 2020).

Once consensus was achieved, a shared coding framework was applied to the full dataset, and iterative refinement continued until no substantially new 1st- or 2nd-order categories emerged, indicating

**Table 3.** Interview themes and sample questions – Source: authors' own work.

| Theme | Example Question | Rationale |
|---|---|---|
| Background Questions | To set the stage, can you briefly describe your current role and responsibility? | To understand participants' organizational role and relevance to GAI-related strategic decision-making. |
| AI Adoption and Strategic Decision-Making | Could you walk me through a typical strategic decision-making process in your organization? | To explore how SDM is currently practiced and where GAI plays a role, if at all. |
| Human-AI Collaboration | Which aspects of SDM are handled by humans and which by AI-driven tools? | To uncover perceptions of responsibility, collaboration, and value-sharing between human actors and GAI. |
| Implementation and Organizational Impact | What changes did the introduction of GAI in SDM entail (process design, efficiency, work organization, etc.)? | To examine the structural and procedural implications of integrating GAI into established decision workflows. |
| Future Outlook and Closing | Where do you see the biggest potential for AI in decision-making? | To capture expectations about future developments and long-term GAI impact on strategic decision-making practices. |

**Table 4.** Overview of collected data – source: authors' own work.

| Data Source | Number | Minutes | Pages | Word Count | Description |
|---|---|---|---|---|---|
| Interviews | 27 | 664 | ./. | ./. | Semi-structured and recorded interviews |
| Internal Presentations/Policies | 6 | ./. | 65 | ./. | Internal presentations and policies concerning GAI |
| Internal Trainings | 2 | ./. | 20 | ./. | Internal online trainings for AI usage |
| Company Report | 1 | ./. | 93 | ./. | Official company reports, such as non-financial annual reports |
| News Articles | 13 | ./. | ./. | 13,380 | Article from company blogs, press, and industry analysis |
| Industry Reports | 5 | ./. | 158 | ./. | External industry reports concerning AI/GAI for luxury sector |

a theoretical saturation (Corbin & Strauss, 2015). To further enhance credibility, cross-validation sessions were conducted among the authors, and member checks were performed with selected interviewees to validate the plausibility of interpretations and ensure that the analytical representation resonated with organisational realities.

The resulting data structure is presented in Figure 1. To further enhance analytical transparency and interpretive rigour, we provided a comprehensive evidence-to-concept mapping in the Appendix (Table A2). This table systematically links the identified design dimensions, 2nd-order themes, and 1st-order concepts with representative illustrative quotes, interviewee sources, and interpretive explanations. By explicitly showing show empirical insights were abstracted into conceptual categories, Table A2 strengthens the traceability of our analytical reasoning and the overall credibility of the findings.

Throughout the coding process, we iteratively refined and consolidated our coding scheme as new insights emerged and conceptual clarity developed. Early codes were frequently merged, split, or relabelled to more precisely reflect the mechanisms underlying GAI integration. For example, preliminary codes such as 'AI ambiguity' and 'Reliability and accuracy concerns' were grouped under the broader theme 'Trust and Confidence' once recurring patterns around reliability and interpretability became evident. In contrast, the initial umbrella category 'Implementation Challenges' was differentiated into more specific themes such as 'Prompt Engineering Deficiency' and 'Data Readiness for GAI' to separate technical barriers from organisational enablers. Challenging coding cases, such as whether statements about human validation of GAI outputs belonged under 'Oversight and Final Decision' or 'Collaboration & Augmentation', were resolved through iterative memoing and collective discussion among the authors. This reflexive, consensus-oriented approach ensured conceptual coherence and interpretive alignment, thereby enhancing transparency and ensuring that the resulting data structure accurately captured both the empirical material and the socio-technical framing of the study. Coding continued until no substantial new 1st- or 2nd-order categories emerged, indicating that theoretical saturation had been reached. This provided confidence that the data structure comprehensively represented the core dynamics of GAI integration in strategic decision-making.

## 4. Findings

This study investigates how GAI is integrated into strategic decision-making (SDM) within a complex organisational setting. Drawing on insights from a single-case study, we explore how GAI-generated outputs are embedded across key phases of the decision process, which are identified from early problem scoping and alternative generation to evaluation and execution. Rather than replacing human judgement, GAI acts as a decision-support layer that enhances analytical capabilities, accelerates reasoning, and expands the scope of considered options. Human actors retain oversight and apply contextual interpretation to ensure that AI-generated insights align with broader strategic objectives. Consistent with our findings, we interpret the observed changes as incremental and bound by governance rather than transformative. Our claims are limited to admissibility, routing, and the allocation of roles between humans and AI within decision episodes.

Grounded in a qualitative, inductive research design, our findings reveal a multi-dimensional integration process shaped by technological, organisational, and human factors. The resulting framework (Figure 2) synthesises three interrelated layers: (1) critical enablers and barriers that condition GAI adoption, (2) the evolving dynamics of human-AI interaction in decision-making contexts, and (3) a four-stage integration and operationalisation process, ranging from early experimentation to institutionalisation.

This framework was directly derived from our empirical coding structure (Figure 1). First, we identified 1st-order concepts based on informant-centric statements. These were then iteratively clustered into 2nd-order themes that reflected recurring mechanisms, tensions, and routines related to GAI integration. In a final abstraction step, we grouped these themes into higher-level design dimensions, which served as the foundational elements of the socio-technical framework. Each component of the model reflects an inductive synthesis of these dimensions, structured to represent both static conditions (e.g. enablers, barriers) and dynamic processes (e.g. integration stages, interaction patterns). Therefore, we report patterns that are consistent with admissibility and routing mechanisms, rather than general performance outcomes.

Throughout this process, we moved iteratively between the empirical material, theoretical constructs (e.g. STS, HITL, XAI), and conceptual abstraction to ensure analytical coherence and grounding. The result is

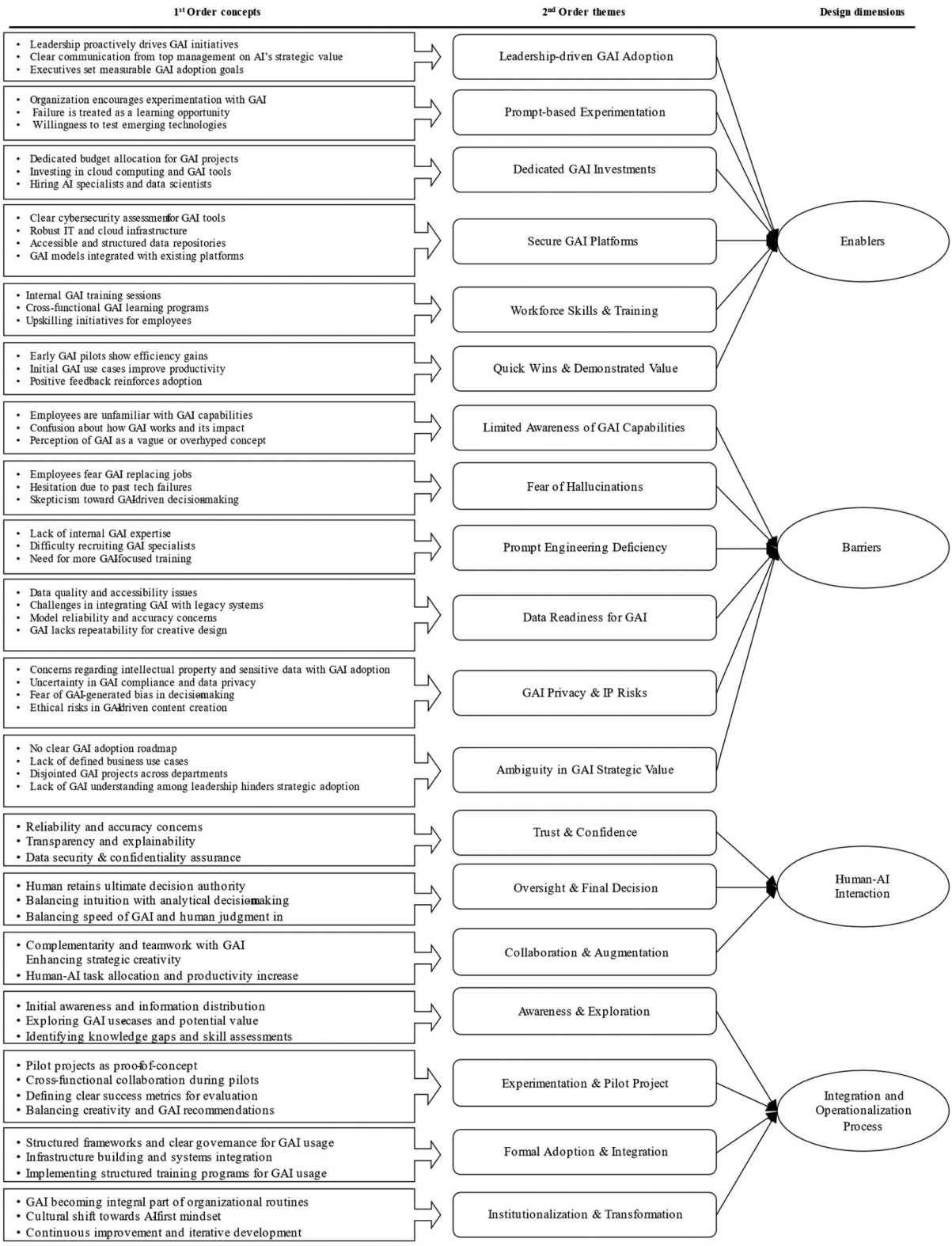


**Figure 1.** Data structure – Source: Authors´ own work.

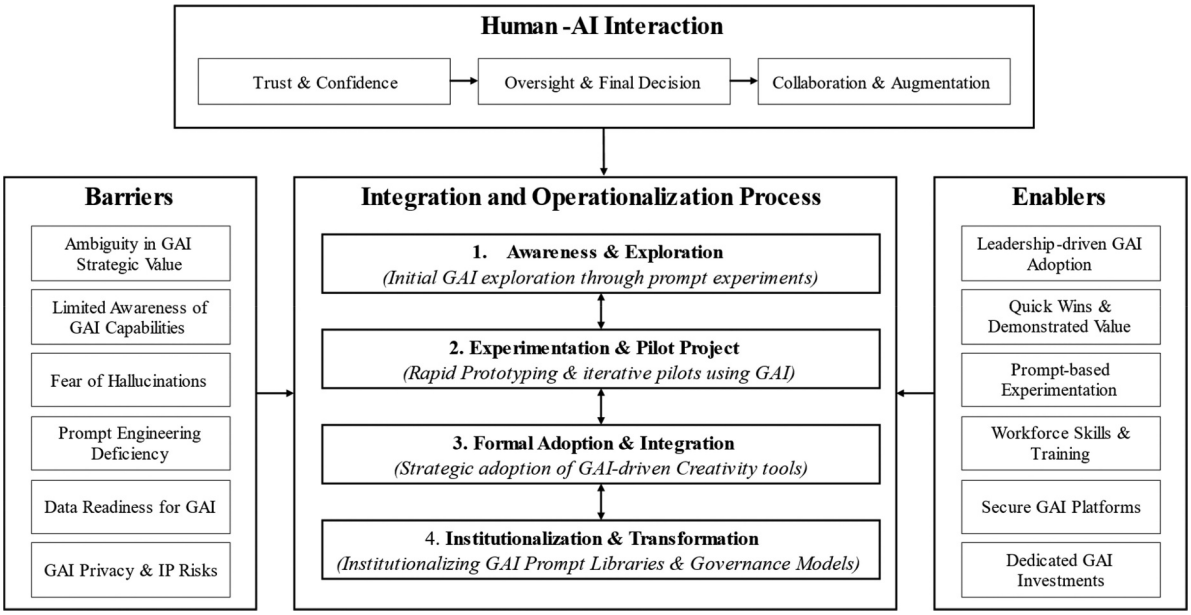


**Figure 2.** A socio-technical framework for GAI integration into strategic decision systems – Source: Authors´ own work.

a data-driven yet theoretically informed framework that captures the socio-technical conditions shaping GAI´s strategic role.

Our results highlight that successful GAI integration is not solely a technical challenge but one of socio-technical system design. Cultural alignment, collaboration routines, and ethical safeguards are essential to realise the potential of GAI in decision systems. In particular, trust, explainability, and organisational readiness emerge as critical prerequisites of adoption. To strengthen the theoretical grounding and broader applicability of our findings, we systematically extended our indicative insights through integration with extant scholarly literature on AI governance, socio-technical decision support, and human-in-the-loop collaboration.

This study explores the practical role of GAI in strategic work by examining how AI-generated outputs are embedded across SDM phases. GAI contributes to early problem identification, supports alternative generation, and assists in evaluating options under uncertainty. Human actors maintain oversight to ensure alignment with strategic objectives. Integrating GAI into planning accelerates analytics and improves clarity and agility in strategic deliberation (Hannila et al., 2022). In our case, integrating GAI into the planning process improves the speed and clarity of analysis in specific situations and broadens the range of options, which is consistent with research on decision support. Therefore, we interpret these patterns as localised, governance-bound changes in decision-making practices rather than as evidence of organisation-wide transformation.

### 4.1. Enablers and barriers

Understanding the conditions that facilitate or hinder the integration of GAI into SDM is essential for building effective decision systems. Based on an in-depth, inductive analysis of executive interviews, our findings highlight a set of key enablers and barriers that shape GAI adoption within complex organisational contexts. These factors operate across technological, organisational, and human domains, influencing the extent to which GAI can be effectively embedded into decision-making processes. Identifying and differentiating these elements provides a nuanced foundation for designing governance structures and human-AI collaboration models that ensures trustworthy, scalable integration.

#### 4.1.1. Enablers

Our findings reveal six key enablers that support the effective integration of GAI into SDM. These enablers reflect critical organisational, technological, and human-system conditions that promote readiness,

accelerate learning, and foster trust in the collaborative use of GAI. Each is grounded in inductive empirical data and contextualised through relevant literature on socio-technical systems, AI adoption, and decision support.

*Leadership Commitment Strategic Framing*: Leadership engagement is a decisive enabler in GAI adoption. Senior executives who actively champion GAI signal its strategic relevance, help reduce uncertainty, and legitimise experimentation across units. Their involvement also ensures alignment with core decision-making logics (Denti & Hemlin, 2012; Kumar et al., 2025). As LuxCorp´s CTO noted:

> We strongly believe that we don't do IT for IT, we do IT for business [.] I am a strong defender of AI overall. (R10, CTO, Holding)

Such visible commitment acts as a cognitive anchor that guides organisational attention, lowers resistance, and fosters a culture of responsible innovation (Agrawal et al., 2022; Sony & Naik, 2020). It also helps frame GAI not as a siloed technical initiative, but as a scalable decision-support capability.

*Quick Wins and Demonstrated Value*: Early-stage GAI adoption benefits from quick, tangible results that demonstrate functional value and reduce organisational scepticism. These 'low-hanging fruits' create momentum, validate use cases, and help scale adoption through internal storytelling (Kotter, 2012; Shrestha et al., 2019). As one data leader framed it pragmatically:

> Let's put it this way: whether four people lift a pallet or a forklift does the job – I prefer the forklift. Its faster, easier, and still based on what I input. (A1, Head, WatchA)

Such results play a critical role in developing decision confidence and showing that GAI tools are not only functional, but also usable and controllable in strategic contexts (Ehsan et al., 2021).

*Prompt-Based Experimentation*: Prompt engineering emerged as a vital entry point for experimentation. Through iterative testing of prompts and use cases, teams can rapidly learn how to engage with GAI, generating localised knowledge while mitigating risk (Brown et al., 2020; White et al., 2023). A Head of R&D illustrated this agility mindset:

> It's mainly with the people from my 3D project. We are using everything which is available. we are using it, but it goes fast. [. . .] You use it for one thing, and then the next month, there's something there. [. . .] It's better to be agile. (D1, Head, AccessoriesA)

This enabler aligns with socio-technical system thinking, where user adaptability and modular experimentation serve as precursors to system-level integration (Janiesch et al., 2022; Sony & Naik, 2020).

*Workforce Skills and Training*: Building workforce competence is essential for sustained integration. Our findings how that resistance to GAI often stems not from opposition, but from lack of understand, a finding echoed across recent studies on AI adoption (Kallmuenzer et al., 2025; Kumar et al., 2025; Salhab et al., 2024). As one global director noted:

> What I don't understand, I initially reject. [. . .] And the resistance doesn't come from "I don't want this tool", but rather from "I don't understand this tool". And I think that's where we need to start. (I4, Global Director, WatchB)

Training programs, peer learning, and internal champions can bridge this gap and foster trust in human-in-the-loop (HITL) configuration of GAI use (Glikson & Woolley, 2020; Kallmuenzer et al., 2025; Miller, 2019).

*Secure GAI Platforms*: Trust in the platform environment is a foundational enabler. Organisations need secure, auditable environments that address privacy, compliance, and explainability concerns, especially when GAI is used in strategic contexts (Raisch & Krakowski, 2021; Salhab et al., 2024). As LuxCorp´s CISO underscored:

> But in parallel, we have also established our own environment to ensure secure handling of AI and GAI topics. [. . .] and we feel a responsibility towards our employees to provide them with something from which they can obtain reliable information. (R4, CISO, Holding)

Robust platform governance strengthens confidence in outputs and mitigates risks of hallucinations, data leakage, and IP violations (Davenport & Mittal, 2022; Janiesch et al., 2022).

*Dedicated GAI Investments*: Finally, sustained GAI adoption requires explicit resource allocation, including dedicated budgets, business cases, and internal accountability structures. This signals strategic intent and

enables the move from ad hoc experimentation to scaled operationalisation (Krakowski et al., 2023; Rana et al., 2024). As the CTO described:

> And, of course, first of all, it starts with an idea. An idea needs to then develop into a business case. And out of this business case, we need always to have a clear return on investment, and a clear budget associated to that. (R10, CTO, Holding)

Such investments create a stable foundation for long-term capability building and institutionalisation.

#### *4.1.2. Barriers*

Our analysis revealed six critical barriers that inhibit the effective integration of GAI into SDM. These challenges span cognitive, technical, and governance dimensions, and were derived through inductive coding of interviews with senior decision-makers. Each barrier reflects not only practical concerns but also systemic misalignments between technological potential and organisational preparedness. Together these barriers underscore the socio-technical complexity of embedding GAI into high-stakes, human-centred decision systems.

*Ambiguity in GAI Strategic Value*: Uncertainty about how GAI aligns with overarching business objectives can significantly hinder commitment and resource allocation. When the strategic value of GAI remains undefined, initiatives often lack direction and are perceived as experimental rather than mission-critical (Birkinshaw, 2018). This vagueness limits executive sponsorship and slows organisational uptake:

> So far, there hasn't been a clear statement on how we should handle this in the future, whether we should actively use it or not. Therefore, it feels somewhat aimless from that perspective. (M1, Manager, AccessoriesB)

*Limited awareness of GAI Capabilities*: A lack of shared understanding across hierarchical levels constrains opportunity recognition and coordinated implementation. Non-technical stakeholders often struggle to grasp GAI´s potential or assess its limitations, which weakens cross-functional collaboration and reinforces siloed experimentation (Castelli & Manzoni, 2022):

> So, I would say it's a balance between the two of making sure that the demos are well documented and well explained to the correct people such that they understand the potential, they are aware of the difficulties of the technology, and they are aware of what it can be used and what it cannot be used for. (R3, Manager, Holding).

*Fear of Hallucinations*: Concerns about erroneous or fabricated outputs remains a core barrier to GAI adoption in high-stakes decision-making contexts. Hallucinations erode trust and raise reputational, legal, and operational risks, particularly when decisions require high levels of precision and accountability (Floridi & Chiriatti, 2020): One senior executive stressed:

> And it's important because until you've addressed some of the key concerns with regards to safety, certainty, [. . .] It hallucinates. It gives you bad information. It gives you bad results. So, I think it's important that we're cautious about how we integrate it with our business to start with. (R11, Director, Holding)

*Prompt Engineering Deficiency*: Effective GAI use depends heavily on users´ ability to formulate precise and structured prompts. However, many employees lack the technical literacy and cognitive framing skills necessary to extract high-quality outputs. Poor prompting leads to irrelevant or misleading insights, undermining perceived usefulness and reliability (White et al., 2023). A digital manager noted:

> And it's also how you ask the questions or the prompting, for instance, because it's not only the way you ask it, but you can also influence the answer based on the questions. and that's really something if you are not clear on what you are asking for sure what you will get will be a bit irrelevant. V1, Manager, WatchD).

*Data Readiness for GAI*: Even the most sophisticated models cannot compensate for poor data inputs. Many organisations face challenges with fragmented datasets, inconsistent data quality, or insufficient access to domain-relevant training material. These constraints limits GAI´s effectiveness and highlight the need for robust data governance strategies (Davenport et al., 2020). A head of analytics explained:

> I talk about again giving to AI the right weight in the answer they provide to you for instance if you have the chatbot and you ask a question for which he doesn't have enough data will not be able to provide you an accurate answer, but the problem is not about AI, the problem is that the input you provide to the systems. (R7, Head, Holding)

*GAI Privacy and IP Risks*: Legal uncertainties around intellectual property, copyright, and data privacy further complicate GAI integration, particularly in creative and marketing domains. Organizations remain cautious due to the lack of clear regulatory guidelines and ownership rules for AI-generated content. As the Chief Legal Counsel warned:

> So, the biggest challenge for AI data input is: are you infringing someone else's copyright or intellectual property? [. . .] And then ultimately the data output risk from an IP perspective is if I'm using Gen-AI to develop a marketing campaign, who owns the copyright in that marketing campaign? If I use Gen-AI to design a new watch, who owns the copyright or patents or whatever?. (R6, Chief Legal Counsel, Holding).

## 4.2. Human-–AI interaction

Beyond macro-level such as leadership, infrastructure, and organisational readiness, the integration of GAI into SDM hinges on the quality of human–AI interaction. This interaction constitutes the micro-level interface where decisions are co-constructed, validated, or contested between human judgement and machine-generated input.

Our findings indicated that successful GAI adoption in decision systems depends on three interrelated dynamics: (1) trust and confidence, (2) human oversight and final decision authority, and (3) collaboration and augmentation. These dimensions shape how human actors relate to the AI system, the extent to which they rely on it, and how it ultimately influences strategic outcomes.

### 4.2.1. Trust and confidence

Trust is the foundational enabler for using AI-generated insights in decision-making. Without it, even technically sound outputs may be rejected or ignored. Our study shows that trust in GAI is formed gradually, through repeated and context-sensitive interactions where users develop a sense of reliability, transparency, and fit-for-purpose utility (BaniHani et al., 2024; Buijsman, 2024).

Participants consistently emphasised the importance of transparency and explainability in fostering confidence (Haupt et al., 2025; Schmidt et al., 2020). Outputs that were not easily interpretable, or whose origin was opaque, were met with scepticism, even when technically accurate (Haupt et al., 2025; Huynh & Aichner, 2025; Raisch & Krakowski, 2021). In contrast, when GAI systems provided clear references, visualisations, or iterative improvements, user trust increased substantially.

This aligns with research highlighting that trust in AI is shaped by perceived competence, transparency, and fairness (Deric et al., 2024; Huynh & Aichner, 2025; Raisch & Krakowski, 2021). Additionally, Ahn et al. (2021) suggest that outcome feedback (seeing whether AI recommendations lead to desirable outcomes) often builds more trust than interpretability alone. In our study, participants noted how successful, low-risk applications, such as internal content generation, acted as 'trust-building playgrounds' before GAI was applied to more strategic topics.

Furthermore, recent findings from the decisions system literature suggest that explainable AI (XAI) is essential in high-stakes contexts, where auditability and validation are crucial (Kleinberg et al., 2020; Ribeiro et al., 2016). This was echoed in our data, where trust required not only model performance, but also confidence calibration, an understanding of when and why the system might fail. As LuxCorp´s CTO put it:

> And I think this is really one of the dangers, and we need to be able to, in the future, to have some sort of guardrails or some sort of mechanism in place that actually measure those answers and are able to either provide a sense of confidence. So a sense of guarantee that the answer is can be trustworthy. (R10, CTO, Holding)

In sum, trust is not a static feature of AI systems but a socio-cognitive process shaped by system behaviour, user experience, organisational culture, and governance mechanisms.

### 4.2.2. Oversight and final decision

While GAI can enhance decision-making by processing large datasets, generating alternatives, and uncovering latent patterns, our findings reaffirm that humans retain ultimate decisions authority, particularly in strategic, high-impact contexts.

Participants consistently articulated that AI should be used to support, not substitute, human judgement. Strategic decisions often involve intangible considerations: implicit knowledge, organisational value, ethical trade-offs, or stakeholder implications that GAI cannot fully model.

This resonates with literature on human-in-the-loop governance models, which emphasise that humans must remain accountable actors in decision systems (Ehsan et al., 2021; Jarrahi, 2018; Shrestha et al., 2019). Oversight acts as a buffer against over-reliance, ensuring that AI outputs are reviewed critically and contextualised appropriately.

Participants employed metaphors to describe this balance. One likened GAI to a 'recipe book', useful for guidance, but ultimately requiring the intuition and experience of a 'chef' to make a good dish. Others noted that retaining control increases users´ confidence and encourages thoughtful engagement with AI suggestions. One R&D manager explained this balance through a vivid analogy:

> So, an AI tool is very helpful, of course, but it does replace human emotion. I can take example for a recipe. When you cook something, you can follow the recipe. And the recipe for me is an AI tool. But when to have a good cooking, to have a good taste, that needs one experience of the cooker. (J1, Manager, WatchC)

Oversight mechanisms observed in the case study included clear escalation pathways, peer review of AI-generated recommendations, explicit human sign-off points in decision workflows, and guardrails limiting GAI use to specific tasks. These practices reflect a broader socio-technical design ethos: AI should be embedded in ways that augment – not automate – human decision roles.

#### *4.2.3. Collaboration and augmentation*

The third key dynamic is the extent to which human decision-makers and GAI systems engage in collaborative sensemaking. Rather than treating GAI as a static tool or oracle, our participants described using it as cognitive partner, a means of expanding thinking, testing assumptions, and generating communicable insights.

This role reflects an augmentation perspective (rather than automation), where GAI serves to enhance clarity, creativity, and reasoning speed. In our data, many executives used GAI not to 'make' decisions directly, but to summarise their own thoughts, explore counterfactuals, test narratives for communication, and draft strategic messages. As one Head of R&D explained:

> Most of the time, actually, I would use it to summarize my own thoughts about a specific problematic . . . the impact in the decision making of the project is not direct . . . expressing my ideas in a more clear way. (D1, Head, AccessoriesA)

This mirrors the logic of hybrid intelligence systems, where AI handles complexity and humans bring contextual interpretation, creativity, and responsibility (Ransbotham et al., 2022). Research also shows that co-creation frameworks, where humans iteratively prompt, review, and refine AI outputs, lead to higher decision quality and user satisfaction (Zhang et al., 2024).

In our case, collaborative augmentation manifested through back-and-forth prompting to refine ideas, using GAI for scenario planning under uncertainty, and applying GAI as a 'thinking assistant' during strategic workshops. However, collaboration also required role clarity, users needed to understand what AI could and could not do, and where human judgement remained indispensable.

### *4.3. Integration and operationalisation process*

The integration of GAI into SDM follows a structured yet iterative process, shaped by organisational context, leadership vision, and technological readiness. Based on our empirical findings, we identify four interrelated stages of operationalisation: (1) Awareness and Exploration, (2) Experimentation and Pilot Projects, (3) Formal Adoption and Integration, and (4) Institutionalization and Transformation.

Each stage reflects a shift in how GAI is perceived and utilised, from isolated experimentation to embedded capability, bringing unique challenges and governance demands.

#### 4.3.1. Awareness and exploration

The process typically begins with an exploratory phase where early adopters within the organisation initiative low-risk, curiosity-driven interactions with GAI tools. These interactions are often decentralised and driven by innovation-orientated units, such as digital labs, transformation teams, or IT leadership, and involve information testing through publicly available platforms or internal sandboxes.

This stage is crucial for building internal momentum. It allows employees to demystify the technology, reduce abstract fears, and being envisioning potential use cases aligned with their own work routines. As noted in the literature, early exposure to AI enables experiential learning and increases the likelihood of long-term adoption by normalising the technology through practical interaction (Janiesch et al., 2022). These initiatives are often driven by innovation-orientated teams and serve to surface viable use cases and prepare internal stakeholders for boarder adoption. As one Head of IT described:

> That means, what we're currently doing as IT is naturally fulfilling our role as innovators and somewhat as those keeping an eye on technological developments in the market. [. . .] We're currently establishing a small AI-champion community that deals with use cases and explores technical possibilities, so that when we start receiving initial inquiries. (A2, Head, WatchA)

This initial phase is also where organisational sensemaking starts, how GAI is interpreted in relation to strategic goals, ethical boundaries, and legacy systems. It sets the foundation for deeper engagement. Such exploration aligns with digital transformation research emphasising the role of structured experimentation in building organisational readiness and linking emerging technologies to strategic priorities (Vial, 2019).

#### 4.3.2. Experimentation and pilot projects

Once awareness is established, organisations typically move into a phase of structured experimentation. This includes pilot projects, internal protypes, and limited-scope rollouts designed to evaluate GAI´s practical utility within defined contexts. These pilots are often championed by transformation teams of domain leaders who act as internal intermediaries between technical and strategic domains.

Participants emphasised that evidence of value is critical at this stage. Tangible outputs, such as faster reporting, higher personalisation, or improved creative quality, build credibility and support a shift from novelty to utility. In line with recent findings (Burger et al., 2023), documenting outcomes and challenges vias 'AI snapshots' can serve as organisational memory, enabling reflective learning. One manager highlighted this iterative, evidence-driven approach:

> We do it first in the group but on specific topic use case that we identified, and we test and we and then we embed it in the organizations. [. . .] That's why we are doing it in the transformation team. Because we provide the proof that it is working and then we integrate it on the day to day. And it's more progressively when you show that things are working, that trust will come as well. (V1, Manager, WatchD)

Governance at this stage begins to take shape through early risk assessments, data access protocols, and lightweight usage guidelines. These early structures are vital for scaling later on.

#### 4.3.3. Formal adoption and integration

In the third stage, GAI moves from pilot experiments to operational deployment, becoming embedded in routine strategic workflows. At this point, the organisation starts treating GAI as a scalable capability, not just a tool. This transition requires aligning GAI with existing enterprise systems, data infrastructures, and decision routines.

Participants described how GAI-generated outputs became a default input into strategic decisions, particularly where data synthesis, scenario planning, or forecasting were required. Integrating GAI into CRM systems, knowledge bases, or workflows tools marks a shift from experimentation to embedded intelligence. This integration involves aligning GAI with enterprise systems, governance frameworks, and existing data infrastructures, thereby enabling more agile and precise decision-making (Asghar et al., 2025; Schwaeke et al., 2024). A marketing director illustrated this operational embeddedness:

> So today we have more than 80 algorithms that are running daily that scores probability to purchase for a prospect, probability to repurchase for an existing customer, and the top three products to be recommended for that customer. And it's fully embedded within our client. (R12, Director, Holding)

Scholars describe this stage as the formation of socio-technical configuration, human and AI actors working within a shared infrastructure governed by protocols, performance metrics, and cross-functional alignment (Dellermann et al., 2022). Trust becomes institutionalised through performance history, governance framework, and cross-team coordination.

#### *4.3.4. Institutionalisation and transformation*

The final stage, institutionalisation and transformation, involves embedding GAI practices into an organization´s long-term strategic orientation, cultural routines, and formal governance structures. At this point, GAI becomes more than a tool, it evolves into an organisational capability anchored in established norms, processes, and strategic reflexivity. We use the term 'transformation' descriptively to denote institutional embedding within decision systems. This phase is typically marked by the creation of governance frameworks, standardised prompt libraries, ethical usage policies, and dedicated roles such as AI councils or champions (Janiesch et al., 2022; Shrestha et al., 2019).

Institutionalisation ensures that GAI adoption remains sustainable, aligned with organisational objectives, and compliant with ethical and regulatory standards. It provides safeguards against misuse, bias, and unintended consequences, thereby supporting responsible innovation and continuous learning. One CTO at LuxCorp underscored the importance of these safeguards:

> So, I think we still need to make sure that whatever AI use case we have, that it respects our AI trustworthy framework. And, that AI is not used to, I don't know, that someone wants to know how to create a bomb, so we will use AI for that. So, I think, yes, it's very important to have those safety nets in place and to ensure that there is a proper usage of those technologies. (R10, CTO, Holding)

This stage reflects recent discussion in decision system literature emphasising AI governance as an adaptive system, combining technical procedural, and cultural safeguards to ensure resilient and ethical use (Dwivedi et al., 2023; Ransbotham et al., 2022). Transformation at this stage is not merely technological. It involves strategic reflexivity, where leaders assess how GAI reshapes their capabilities, decision norms, and value propositions. GAI becomes embedded not only in workflows but also in how the organisation thinks.

#### *4.3.5. Illustrative vignette: from exploration to institutionalisation*

To illustrate how GAI integration unfolds across organisational levels within LuxCorp´s decentralised multi-brand ecosystem, we present a vignette tracing the evolution of a GAI-enabled product copywriting assistant.

The journey began in the *Awareness and Exploration* phase, where early interactions with generative AI tools were initiated by IT leadership and innovation teams. These were low-risk, curiosity-driven experiments, often conducted in sandbox environments. As one Head of IT at WatchA recalled:

> That means, what we're currently doing as IT is naturally fulfilling our role as innovators and somewhat as those keeping an eye on technological developments in the market. [. . .] We're currently establishing a small AI-champion community that deals with use cases and explores technical possibilities, so that when we start receiving initial inquiries. (A2, Head, WatchA)

In LuxCorp´s decentralised ecosystem, these early explorations unfolded differently across brand units, while the holding company provided overarching guidance and infrastructure. Each brand could initiate local pilots aligned with its strategic context, but scaling or formal adoption required coordination with central governance structures. As one innovation manager explained:

> The second level is more like a process. The second factor is more like a process factor. The tool needs to be approved by the group. It needs to go through the trustworthy AI council. (I1, Manager, WatchB)

This dynamic illustrates how decentralisation fostered experimentation and contextual adaption, whereas centralised oversight ensured compliance and alignment with corporate standards. Local teams could act entrepreneurially within a defined governance perimeter, reflecting the interplay between autonomy and control that characterises distributed ecosystems.

This exploratory work laid the foundation for a targeted pilot in the *Experimentation and Pilot Projects* stage. Motivated by the desire to improve the efficiency and consistency of product descriptions across

global markets, the transformation team launched a pilot project to train GAI on the brands copy tone, product metadata, and campaign goals.

The pilot yielded promising results: GAI reduced briefing time, increased localisation quality, and shortened campaign timelines. These outcomes paved the way for *Formal Adoption and Integration*, in which the assistant was embedded into the brand´s marketing workflow. GAI-generated copy was integrated into the product lifecycle management (PLM) system and reviewed via human-in-the-loop process, ensuring brand fit and compliance.

> We are also going to launch from March onwards a program [Tool Name] . . . we can learn how they can use [Tool Name]in a way that could benefit them, enhance the output but also give them time on strategic activities. (R9, Group Director, Holding)

As the tool´s use scaled, the brand moved into the *Institutionalization and Transformation* phase. The AI assistant became part of the standard content creation process, supported by formal governance mechanisms, including prompt libraries, safety checklists, and oversight from the AI Council. A CTO underscored the importance of ethical safeguards:

> So, I think we still need to make sure that whatever AI use case we have, that it respects our AI trustworthy framework. And, that AI is not used to, I don't know, that someone wants to know how to create a bomb, so we will use AI for that. So, I think, yes, it's very important to have those safety nets in place and to ensure that there is a proper usage of those technologies. (R10, CTO, Holding)

This case illustrates how GAI integration unfolds as a socio-technical process, shaped by iterative learning, cross-functional collaboration, and evolving governance. What began as a sandbox experiment eventually became a routinised organisational capability, embedded in LuxCorp´s infrastructure and strategic routines.

Within this decentralised ecosystem, GAI integration followed a dual logic: brand-level autonomy enabled rapid experimentation and contextual innovation, while central-governance, through mechanisms such as the AI Council, ensured coherence, compliance, and ethical oversight. This interplay between local initiatives and group-level alignment illustrates how human actors and GAI co-evolve across organisational layers to shape strategic choices and transformation. Decentralisation thus emerged not as a barrier but as a structural enabler that balanced innovation speed with governance integrity in LuxCorp´s distributed decision system.

## 5. Discussion

Our analysis shows that the successful integration of GAI into strategic decision-making depends not only on technological affordances but also on socio-organisational alignment, role reconfiguration, and governance structures that enable responsible adoption (Csaszar et al., 2024; Goktas, 2024; Ulrich et al., 2025; Zeiser, 2024). This finding is consistent with broader research on digital ecosystems, which emphasises that governance and coordination mechanisms structure the interplay of distributed actors within interconnected digital environments (Bouncken & Kraus, 2022). Taken together, these insights locate our contribution within decision-systems scholarship and empirically motivate a design-oriented lens on interfaces and checkpoints that admit AI outputs into formal deliberation (Heilig & Scheer, 2023; Rodriguez-Garcia et al., 2023; Seeber et al., 2020). We interpret our findings as incremental, governance-bound changes rather than as a complete transformation. Our inferences concern admission, routing, and role allocation in decision systems, but do not claim to cause outcomes.

Furthermore, similar to Held et al. (2025), our findings underline that cultivating digital leadership and an innovation-oriented culture acts as a catalyst for socio-technical alignment and sustained capability building in AI-enabled decision systems. These leadership-driven mechanisms mirror the role of dynamic capabilities identified in studies of digital transformation, suggesting that cultural readiness and organisational learning are indispensable for institutionalising GAI within decision architectures. Accordingly, we frame organisational effects as contingent on gate readiness and governance maturity rather than as universal performance gains.

Our findings both corroborate and extend prior work on AI-enabled decision systems. First, they confirm that explainability, trust, and the embedding of AI into organisational routines are central drivers of organisational uptake, and that HITL accountability remains pivotal in high-stakes SDM. At the same time,

our results add a more fine-grained perspective by introducing a four-gate admissibility logic focused on the entry of GAI outputs into decision forums (quality, provenance, explainability, accountability) and a staged, practice-grounded process model that traces how pilots evolve into routinised inputs within a decentralised ecosystem.

### 5.1. Positioning within decision systems

Beyond domain-specific insights, our findings also contribute to the broader design and evaluation of decision systems by foregrounding a HITL perspective and a socio-technical understanding of GAI integration. Empirical studies indicate that explanation and routine embedding are decisive for organisational uptake of intelligent-system outputs (Ehsan et al., 2021; Heilig & Scheer, 2023; Rodriguez-Garcia et al., 2023; Seeber et al., 2020). We conceptualise GAI not as an autonomous oracle but as decision support that augments managerial judgement across the phases of SDM (problem structuring, option generation, evaluation, and commitment). In line with decision-systems scholarship, we explicate the interfaces, checkpoints, and accountability regimes through which AI-generated outputs become admissible inputs to formal deliberation, thereby specifying procedural 'gates' for quality, provenance, explainability, and accountability. These mechanisms are documented in our case; any broader generalisation is treated as a boundary-aware proposition.

Empirical evidence supports this positioning. Studies show that the provision of explanations and interactive oversight mechanisms critically shape whether outputs from intelligent systems are trusted and adopted in decision forums (Bucinca et al., 2020; Gregor & Benbasat, 1999). Case-based research further demonstrates that AI becomes effective only when embedded into organisational routines and governance structures, highlighting the socio-technical co-evolution of decision systems (Heilig & Scheer, 2023; Rodriguez-Garcia et al., 2023). Collaboration studies emphasise that machines are not merely tools but can function as teammates in collective decision processes, requiring clear role allocation and accountability mechanisms (Seeber et al., 2020). Together, these findings substantiate our interpretation that GAI integration must be conceptualised as a socio-technical and HITL arrangement, not as a purely technological intervention. Beyond the present case, speculative implications include the following: if replicated, gate thresholds will likely tighten with higher stakes and centralisation and relax with social transparency.

This perspective contributes a process model of decision systems that specifies HITL arrangements, admission criteria for AI outputs, and governance templates that convert pilots into routinised decision support. Recent evidence also indicates that organisational users filter GAI recommendations through trust, explainability, and perceived appreciation, and that contextual risk sets admissibility thresholds (Huynh, 2024), which aligns with our gate logic of quality provenance, explainability, and accountability. Therefore, we position our process model as an empirically grounded description of admissibility rather than as a claim about general performance effects.

### 5.2. Theoretical contributions

This study advances research on AI integration into strategic decision making by proposing a comprehensive, process-oriented framework (see Figure 2) for embedding GAI in organisational decision systems. Grounded in our case evidence and prior empirical work, we show how enablers, barriers, human – AI interaction, and a four-stage operationalisation cycle co-evolve in complex multi-actor settings (Jarrahi, 2018; Seeber et al., 2020; Shrestha et al., 2019). We conceptualise these conditions as admission gates that determine the strategic admissibility of AI outputs. Our gate constructions are grounded in empirical evidence and should be considered boundary-aware propositions for future testing. Prior studies support this logic. For example, Kim et al. (2020) link opacity in black-box models to reduced trust and use, emphasising the importance of explainability and accountability. Additionally, Shin (2021) demonstrates that 'causability' and the quality of explanations predict trust and acceptance. Furthermore, Saffarizadeh et al. (2024) identify creator-based alignment as a socio-technical proxy for provenance. These findings collectively support the conceptualisation of admission gates as threshold conditions for the use of AI outputs in human decision-making, especially in high-stakes settings. They also align with our identification of quality, provenance, explainability, and accountability as necessary gate criteria.

Against this backdrop, our theoretical contributions unfold along three main lines.

First, we specify a socio-technical alignment problem in which analytical affordances of GAI outpace governance logics in distributed decision systems, and we resolve this tension by identifying admission gates and showing how HITL arrangements operationalise responsibility across decision phases.

Second, we synthesise organisational enablers (Leadership driven GAI Adoption; Quick Wins & Demonstrated Value; Prompt-based Experimentation; Workforce Skills & Training; Secure GAI Platforms; Dedicated GAI Investments) and adoption barriers (Ambiguity in GAI Strategic Value; Limited Awareness of GAI Capabilities; Fear of Hallucinations; Prompt Engineering Deficiency; Data Readiness for GAI; GAI Privacy & IP Risks) and show their dynamic coexistence (Durmusoglu et al., 2022; Fichman & Kemerer, 1999; Glikson & Woolley, 2020; Kotter, 2012).

Third, we contribute operationalisation by treating gates as measurable design variables rather than metaphors. Building on Huynh (2024), trust formation mechanisms and explainability features function as gate levers, yielding testable propositions on when outputs transition from pilots to admissible decision inputs. We argue that admissibility follows a sequence in which the value of later gates (explainability, accountability) is conditioned by early gates (quality, provenance), producing distinctive failure patterns and clear leading indicators.

Moreover, our framework positions human-AI interactions as key mediators that shape decision quality. By foregrounding hybrid decision making, we highlight the evolving role of human actors (Patil, 2024; Raisch & Krakowski, 2021) alongside the analytical capabilities of AI. Our findings emphasise that organisational culture, trust, oversight, and transparency remain central to sustainable AI integration (Haupt et al., 2025; Jaboob et al., 2025; Kartikeya, 2022; Miller, 2019; Wohleber et al., 2021; Yaprak, 2024). While the analysis primarily foregrounds enablers and governance mechanisms that facilitated GAI integration, our data also reveal the relative absence of overt resistance or power struggles. This pattern likely reflects the strong top-down framing of GAI as a strategic initiative, which channelled potential dissent into formal governance routines rather than open contestation. We interpret this managerial framing as both a strength and a limitation: it enabled alignment and responsible adoption, yet may have muted critical reflection on distributional effects, autonomy, or potential dependency on algorithmic outputs. Recognising this tension adds a critical reflection to our framework and invites future research to examine how technological optimism and control logics interact in shaping GAI integration trajectories (Janssen, 2025; Stohr et al., 2024).

Finally, our four-step process Awareness, Experimentation, Formal Integration, Institutionalisation provides a structured roadmap for researchers and practitioners and addresses existing gaps in AI implementation in complex business ecosystems (Bag et al., 2021; Fichman & Kemerer, 1999). By emphasising incremental learning, iterative prototyping, and cultural assimilation, our framework directly responds to calls for detailed insights into technology diffusion in industrial contexts.

Overall, the framework reinforces the relevance of socio-technical systems theory, which emphasises the co-evolution of technical infrastructure and cultural readiness (Geels, 2004; Sony & Naik, 2020). Focusing on these socio-technical dynamics provides an empirically grounded understanding of the systematic embedding of AI in strategic decision making. This provides a foundation for future research on the scalable, ethical, and trustworthy use of AI in different organisational settings.

### 5.3. Practical implications

Our framework provides guidance for strategic decision-makers, senior executives, and functional leaders across organisational settings, and foregrounds evidence-based levers, quick wins, secure platforms, targeted training, and iterative experimentation, for adoption (Burger et al., 2023; Davenport et al., 2020; Janiesch et al., 2022; Kotter, 2012; White et al., 2023).

Companies should simultaneously establish clear ethical guidelines and governance structures to build trust and manage operational and reputational risks effectively, for example through interdisciplinary oversight committees (Bevilacqua et al., 2025). In high-stakes contexts, provenance and explainability act as gating criteria for strategic admissibility, aligning with observed trust/oversight dynamics in human – AI collaboration (Ehsan et al., 2021; Miller, 2019; Shrestha et al., 2019).

In the mid-term, organisations should actively promote experimentation and iterative pilot projects to validate GAI use cases in controlled settings (Sin & Kathiarayan, 2023). This iterative approach helps reduce

implementation risks and strengthens internal confidence. Efforts at this stage must align GAI initiatives with existing operational processes, possibly requiring workflow adjustments, resource reallocation, and clearly defined roles (e.g. 'AI Champions') to facilitate integration and organisational learning (Davenport et al., 2020).

Over the long-term, companies must institutionalise formal GAI governance frameworks embedding ethical standards, security protocols, and compliance mechanisms into their organisational structures. Sustainable integration requires continuous leadership commitment and substantial investments in technology, skills development, and adaptability (Bevilacqua et al., 2025; Schwaeke et al., 2025). As demonstrated by Held et al. (2025), digital transformation succeeds where digital leadership and a participative culture jointly foster trust and experimentation; similarly, GAI integration benefits from governance designs that balance autonomy and accountability within distributed ecosystems (Bouncken & Kraus, 2022). Long-term success hinges not only on technical excellence but also on fostering a resilient, learning-oriented culture that continuously evolves alongside AI capabilities (Kallmuenzer et al., 2025).

Managers should treat gate conditions as design parameters of decision systems, not as after the fact checks. Concretely, provenance requires source citation in prompts and outputs by default; explainability benefits from layered rationales with drill down; accountability depends on explicit human approval points; trust can be increased by minimising sensitive inputs and providing user control over disclosure (Huynh, 2024). Near term metrics include the share of decisions with attached provenance, explanation coverage and time to understanding, the share of decisions with a named owner, user trust scores and perceived benefit risk balances; outcome proxies include post hoc error rates, rework, escalation frequency, time to decision, compliance findings, and performance against ex ante criteria. These levers translate into governance templates and dashboards that render gates auditable at the point of deliberation.

Empirically, our data reveal four recurrent failure modes when scaling along the pathway. First is gate superficiality, where explanations and provenance exist but are not invoked in deliberation. This yields low decision usefulness, which is consistent with evidence showing that acceptance depends on the use of explanations and social transparency rather than mere availability (Ehsan et al., 2021; Miller, 2019). Second is role drift in HITL, where sign-offs diffuse across functions and escalation slows. This echoes the risks of socio-technical coordination in governance settings (Heilig & Scheer, 2023; Seeber et al., 2020). Third is metric myopia, where usage increases while outcomes remain stagnant. Aligning usage with decision-quality indicators improves the signal, which is consistent with decision analysis guidance (Montibeller & Franco, 2010) and trust and benefit pathways (Huynh, 2024). Fourth, pilot habituation: PoCs persist without admission to decision forums, reflecting known organisational barriers to scaling (Bag et al., 2021; Rana et al., 2024). In our case, mitigation correlated with treating the four gates as design parameters, assigning single-point accountability per episode, and tracking gate-aligned outcomes (e.g. share of decisions with provenance, rework/error rates, and time to decision). These implications are practice-oriented and do not imply causal performance effects.

### 5.4. Limitations and future research

This study uses a qualitative single case design (Yin, 2018) and an inductive analytical approach following a structured coding procedure in the sense of Gioia et al. (2013) to examine how GAI is integrated into organisational decision systems. While this yields rich contextual insight, several limitations remain. In line with theory-building case research, a single-case design is appropriate when constructs are only partially specified and the setting is revelatory. In this case, external validity is analytic rather than statistical (Yin, 2018). Consequently, we present our claims as theoretical propositions open to examination in contrasting contexts.

Evidence comes from a single anonymised multi brand group with decentralised structures; transferability may be constrained by governance, data and capability profiles. Replication across sectors with different regulatory intensity and decision criticality, and across governance regimes with varying central oversight, is needed. Our inferences are influenced by decentralised governance. In highly centralised organisations, the importance, timing, and ownership of the mechanisms we report may differ, suggesting that the mechanisms become less significant as decision-making authority becomes more centralised.

Data rely primarily on semi structured interviews and documents, with risks of self-report, recall and social desirability. The absence of direct observation and digital trace data limits triangulation. Future work should add non-reactive materials such as meeting minutes, workflow and ticket logs, prompt logs and AI snapshots, model usage telemetry, decision artefacts, approval histories, and post decision performance indicators. Inductive coding produced first-order concepts, second-order themes and aggregate dimensions, but no formal inter coder reliability statistics were reported. Preregistered codebooks, inter coder agreement and transparent audit trails would strengthen dependability.

Purposeful sampling targeted participants directly engaged in GAI-related initiatives, governance, or strategic planning, which may have led to an overrepresentation of advocates and early adopters. While this focus ensured insight into the mechanisms of implementation it may have limited the inclusion of more sceptical or less engaged perspectives. We attempted to balance this by including participants from both functional and governance roles; however, access constraints and the emergent nature of GAI initiatives made it difficult to systematically include dissenting voices. Future research should therefore incorporate maximum variation and negative case sampling, explicitly seeking out units or individuals who discontinued pilots, resisted adoption, or questioned the organisational value of GAI to capture a fuller spectrum of perspectives.

Relatedly, the limited presence of resistance, power asymmetries, or unintended consequences in our findings may stem from the sample composition and organisational context. As participants were predominantly involved in implementation and leadership roles, narratives of alignment and progress were more visible than those of contestation or scepticism. We therefore caution that the absence of critical perspectives should not be interpreted as their non-existence but as a reflection of the governance maturity and framing of GAI initiatives in this case. Future research should explicitly investigate resistance, politics, and unintended effects, such as the displacement of human judgement or reinforcement of hierarchical control as integral components of socio-technical GAI integration.

The process model is explanatory, not causal. It does not identify whether gates for provenance, explainability and accountability improve decision quality or speed, nor binding thresholds. Quasi experimental and experimental designs are needed, for example stepwise rollouts of governance templates, interrupted time series around policy changes, field experiments that vary explanation depth or provenance displays, and difference in difference designs where units adopt secure GAI platforms at different times.

Data capture an early, cross-sectional window; capabilities, platforms and policies evolve quickly. Longitudinal and multistage designs should trace trajectories from exploration and pilots to integration and institutionalisation, including reversals and failure cases.

Trust, explainability, provenance and accountability were theorised qualitatively. Future studies should combine validated psychometric scales with behavioural and log-based measures, for example citation rates in prompts, share of decisions with provenance attached, escalation frequency, time to decision, and outcome proxies such as post hoc error rates, rework, compliance findings and performance against ex ante criteria. For multi objective episodes, Multi Criteria Decision Analysis can be coupled with explanation metrics to assess admissibility thresholds.

Additionally, a comparative case design that contrasts decentralised firms with highly centralised, process-driven corporations and public-sector agencies operating under statutory constraints would directly test the boundary conditions of our framework. This design would also examine whether the proposed mechanisms travel across governance regimes.

Anonymisation limits sharing of primary materials. De identified codebooks, excerpt tables, protocol templates and analysis scripts should be published where permissible; registered reports can increase procedural transparency.

To address these gaps, we recommend a mixed method program that combines comparative case studies across governance regimes, longitudinal process tracing of decision episodes and field experiments embedded in decision workflows. Design science efforts could prototype decision gate dashboards that make provenance, explanation and accountability auditable at the point of deliberation and evaluate their effects on decision outcomes. We position our contributions as empirically grounded propositions, not universal claims. They invite replication and refinement in structurally distinct contexts.

## 6. Conclusion

This study examined how GAI is embedded into SDM within complex, multi actor organisational settings. Based on a single case study enriched with additional data sources, we developed a framework that identifies key enablers and barriers, outlines human-AI interaction dynamics, and details a four stage GAI integration process.

Our findings show that successful GAI adoption depends on leadership commitment, targeted investment, demonstrable value creation, and internal capability building. At the same time, organisations must address challenges such as data limitations, unclear strategic value, and ethical concerns. Consistent with our evidence, we frame GAI as decision support in HITL arrangements, where ultimate accountability remains with human actors and AI outputs must satisfy explicit admission gates quality, provenance, explainability, accountability to enter formal strategic deliberation.

We reframe GAI as decision support within decision systems and clarify the conditions under which outputs become strategically admissible. Our four-stage pathway: Awareness and Exploration; Experimentation and Pilots; Formal Adoption and Integration; Institutionalisation and Transformation, traces how pilots mature into routinised decision inputs under HITL accountability. By operationalising admission gates and linking them to trust mechanisms evidenced in recent work (e.g. Huynh, 2024), we offer a testable architecture and practical templates for scaling GAI responsibly.

This paper presents an account of how generative AI becomes a routine input in strategic decision-making using a decision-systems approach. Our framework introduces four admission gates: quality, provenance, explainability, and accountability, as decisive preconditions for admitting AI outputs into formal deliberation. This shifts the focus from generic adoption to governed admissibility. We present an empirically grounded process model that traces the transition from pilot programmes to stable decision inputs. This model

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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## Declaration of generative AI in the writing process

The final draft of this manuscript has been proofread regarding grammatical correctness, and fluency by OpenAI's ChatGPT 5 (version October 2025), https://chatgpt.com (last accessed on 9 October 2025). The authors carefully investigated and revisited all proposed changes to ensure integrity and high quality of the submitted manuscript.

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## Appendix

**Table A1.** Semi-structured interview Guide on GAI integration into SDM – Source: Authors´ own work.

| Theme | Sub-Theme | Question |
|---|---|---|
| Introduction | ./. | Introductory questions about background and consent for audio taping. |
| Background Questions | Role and Experience | To set the stage, can you briefly describe your current role and responsibility? |
| | | How involved are you in decision-making related to AI adoption? |
| | | What has been your experience with AI-driven tools or processes so far? |
| AI Adoption and Strategic Decision-Making | SDM Process and Tools | Could you walk me through a typical strategic decision-making process in your organisation? |
| | | How has the approach to decision-making changed in recent years? |
| | | What kind of tools or systems does your company use to support strategic decision-making? |
| | | Can you recall a situation where a new technology helped or changed how decisions were made? |
| | | What kinds of insights or data are most valuable when making strategic decisions? |
| | | When new technologies are introduced in strategic decision-making, what factors drive their adoption? |
| Human-AI Collaboration | Roles and Value | Which aspects of SDM are handled by humans and which by AI-driven tools? |
| | | If AI-generated insights or recommendations are presented, are these insights directly applied, discussed, or further refined by human decision-makers? |
| | | How has/could AI changed the way teams collaborate when making strategic decisions? |
| | | What kind of training or support (if any) would be needed for employees to work alongside with AI? |
| | | Can you describe how decisions are made when AI recommendations conflict with human intuition or expertise? |
| | | What do you think are the biggest challenges in creating a successful collaboration between AI and human decision-makers? |
| Implementation and Organizational Impact | Integration and Challenges | What changes did the introduction of GAI in SDM entail (process design, efficiency, work organisation, etc.)? |
| | | What are the biggest challenges in integrating AI into decision-making in your experience? |
| | | Have you faced resistance from employees or leadership regarding AI adoption? If so, why? |
| | | Are there technical or ethical concerns about AI-driven strategies? |
| | | Which factors were particularly important for the successful implementation? |
| | | What factors influence whether your Maison/department adopts AI-driven marketing tools? |
| Future Outlook and Closing | Outlook | Where do you see the biggest potential for AI in decision-making? |
| | | How do you think the role of human will evolve with AI advancements? |
| | | If you could improve one aspect of AI-driven, what would it be? |
| | Closing | Is there anything else you would like to add or any insights we may have missed? |

**Table A2.** Mapping of empirical evidence to data structure: illustrative quotes, 1st-order concepts, and interpretation notes - Source: Authors´ own work.

| Design Dimension | 2nd-Order Theme | 1st-Order Concept | Illustrative Quote(s) | Interviewee | Interpretation Note |
|---|---|---|---|---|---|
| **Enablers** | Leadership-driven GAI Adoption | Leadership proactively drives GAI initiatives | Our leadership team proactively pushes for GAI integration across departments to create strategic value. | R2 Chief of Staff | Leadership sponsorship legitimises experimentation and ensures strategic coherence across units. |
| | | Clear communication from top management on AI's strategic value | Ambition set by the board of directors to achieve targets – clear leadership involvement. | R11 Director | Visible top-level communication signals organisational commitment and reduces ambiguity. |
| | Prompt-based Experimentation | Organization encourages experimentation with GAI | We started small with prompt testing – low risk, but insightful in how GAI could support us. | M1 Manager | Iterative experimentation fosters learning and low-risk exploration. |
| | | Failure is treated as a learning opportunity | If a GAI pilot fails, we don't see it as a loss – we ask what did we learn from this iteration? | R2 Chief of Staff | Treating unsuccessful pilots as learning opportunities creates psychological safety and reinforces a culture of continuous improvement, turning short-term setbacks into organisational knowledge. |
| | Dedicated GAI Investments | Dedicated budget allocation for GAI projects | There is a defined GAI budget and resources allocated centrally – we don't have to fight for it.'I'm part of our Group AI Council . . . review any new AI demands and allocate funding'. | R10 CTO | Strategic budgeting institutionalises innovation and accountability for resource use. |
| | | Investing in cloud computing and GAI tools | We have secured a multi-year budget for AI integration across functions. | R11 Director | Predictable funding promotes sustained scaling beyond pilots. |
| | Secure GAI Platforms | Clear cybersecurity assessment for GAI tools | We only use GAI platforms approved by our internal IT security – this is a non-negotiable. | R4 CISO | Security compliance increases organisational trust and user confidence. |
| | | GAI models integrated with existing platforms | System integration is critical; new AI solutions must fit our IT landscape. | R6 Chief Legal Counsel | Compatibility with secure infrastructure safeguards data and accelerates acceptance. |
| | Workforce Skills & Training | Internal GAI training sessions | We established a GAI champion network to train and disseminate best practices across brands. | A2 Head | Structured training programs close the knowledge gap and empower employees. |
| | | Upskilling initiatives for employees | We launched an AI curriculum for marketing and innovation teams to build strategic capabilities. | R9 Group Director | Targeted upskilling embeds AI literacy in decision-critical areas. |
| | Quick Wins & Demonstrated Value | Early GAI pilots show efficiency gains | A few early wins helped convince more sceptical teams – they saw the time saved instantly. | V1 Manager | Early performance proof builds momentum and internal legitimacy. |
| | | Initial GAI use cases improve productivity | Early GAI pilots already showed we could shorten content creation time by 60 %. | V1 Manager | Demonstrated productivity gains from initial GAI use cases build tangible proof of value, strengthening credibility and promoting organisational buy-in for broader adoption. |
| **Barriers** | Limited Awareness of GAI Capabilities | Employees are unfamiliar with GAI potential | People still think of GAI as a chatbot – they don't understand how it helps with strategy. | R7 Head | Knowledge gaps slow diffusion and restrict initiative scope. |

(*Continued*)

**Table A2.** (Continued).

| Design Dimension | 2nd-Order Theme | 1st-Order Concept | Illustrative Quote(s) | Interviewee | Interpretation Note |
|---|---|---|---|---|---|
| | | Confusion about how GAI works and its impact | Most colleagues aren't trained yet, which limits adoption of AI solutions. | C2 Senior Manager | Lack of transparent success stories perpetuates misconceptions. |
| | Fear of Hallucinations | Scepticism towards GAI-driven decision-making | There was a case where the GAI output included a fake quote – that really hurt trust. | R1 Manager | Perceived inaccuracy fosters reluctance to rely on AI outputs. |
| | | Hesitation due to past tech failures | We've seen many digital initiatives fail before, so some colleagues are naturally cautious with GAI | R7 Head | Negative experiences with previous technologies reduce risk appetite and slow adoption, indicating the importance of rebuilding trust through small, low-risk GAI trials. |
| | Prompt Engineering Deficiency | Lack of internal GAI expertise | Nobody really knew how to write prompts – it's a skill we overlooked at first. | C1 Manager | Missing prompt literacy leads to poor outputs and user frustration. |
| | | Need for more GAI-focused training | Our teams need deeper training beyond basic prompts – they need to understand what's behind the model"."Right now, awareness is there, but we lack hands-on training for everyday use. | R7 Head | The absence of comprehensive, role-specific GAI training limits user confidence and restricts effective application in strategic decision contexts. |
| | Data Readiness for GAI | Data quality and accessibility issues | Our biggest barrier is legacy data – it's simply not structured enough for GAI use. | R5 Head | Data fragmentation hinders model performance and confidence. |
| | | Challenges in integrating GAI with legacy systems | We still lack cross-functional data linkages that AI requires to learn effectively. | C2 Senior Manager | Disconnected systems restrict scaling and automation. |
| | GAI Privacy & IP Risks | Concerns regarding intellectual property | Legal warned us about potential IP leaks through external GAI tools. | R6 Chief Legal Counsel | Perceived IP and privacy risks deter open experimentation. |
| | | Uncertainty in GAI compliance and data privacy | We need explicit rules on what can be shared with GAI tools. | R2 Chief of Staff | Clarity on boundaries reduces compliance fear. |
| | Ambiguity in GAI Strategic Value | No clear GAI adoption roadmap | Some see GAI as useful only for marketing texts, others expect full strategic support – there's no consensus. | R11 Director | Misaligned expectations lead to fragmented initiatives. |
| | | (A) D | We still don't know who owns AI strategy – IT or business. | R7 Head | Role ambiguity weakens accountability for implementation. |
| **Human – AI Interaction** | Trust & Confidence | Reliability and accuracy concerns | Without clear sources or logic, it's hard to trust GAI recommendations for high-stakes decisions. | C2 Senior Manager | Explainability builds confidence and reduces cognitive dissonance. |
| | | Transparency and explainability | AI needs to be transparent and explainable . . . we explain why the forecast is as it is. | R7 Head | Visible reasoning enhances perceived credibility of AI insights. |
| | Oversight & Final Decision | Human retains ultimate decision authority | Whatever GAI generates still goes through a human – the final call is always ours. | R12 Group Director | Human oversight preserves accountability and ethical judgement. |
| | | Balancing speed of GAI and human judgement in | AI delivers answers in seconds, but we still need time to reflect before making big strategic calls. | R11 Director | Decision-makers must balance the efficiency of GAI with deliberate human judgement to prevent overreliance and maintain accountability in strategic decision-making. |
| | Collaboration & Augmentation | Complementarity and teamwork with GAI | We use GAI to brainstorm and simulate options, but humans still synthesise and decide. | W1 CIO | Complementarity enhances efficiency and idea generation. |

(*Continued*)

**Table A2.** (Continued).

| Design Dimension | 2nd-Order Theme | 1st-Order Concept | Illustrative Quote(s) | Interviewee | Interpretation Note |
|---|---|---|---|---|---|
| | | Enhancing strategic creativity | AI helps us explore different strategic scenarios we might otherwise miss. | C1 Manager | GAI broadens the range of strategic alternatives considered. |
| **Integration and Operationalization Process** | Awareness & Exploration | Initial awareness and information distribution | We started with a tech scouting team exploring what GAI might solve internally. | A1 Head | Early awareness initiates cultural and cognitive readiness. |
| | | Identifying knowledge gaps and skill assessments | We first mapped out who actually uses GAI and where the knowledge gaps are. | A2 Head | Conducting skill audits clarifies capability gaps and guides targeted upskilling, enabling structured capacity building and sustainable GAI integration. |
| | Experimentation & Pilot Project | Pilot projects as proof-of-concept | We tested GAI in one small use case and monitored the effect on campaign design speed. | I2 Associate Director | Pilots bridge conceptual exploration and operational adoption. |
| | | Defining clear success metrics for evaluation | Clear KPIs must be established to assess if AI adds tangible value. | C2 Senior Manager | Metrics turn experimentation into measurable performance insights. |
| | Formal Adoption & Integration | Structured frameworks and clear governance for GAI usage | Now it's fully integrated – GAI outputs automatically flow into our CRM and trigger campaigns. | R12 Group Director | Governance embeds GAI in organisational routines. |
| | | Infrastructure building and systems integration | We're building the infrastructure needed for better data consolidation across teams. | C2 Senior Manager | Systems integration operationalises AI outputs within existing workflows. |
| | Institutionalisation & Transformation | GAI becoming integral part of organisational routines | We now have a GAI ethics board and prompt library – this has become part of our standard processes. | R9 Group Director | Institutionalisation marks maturity – GAI shifts from experimental to constitutive element of SDM. |
| | | Continuous improvement and iterative development | AI institutionalisation involves regular updates to stay aligned with business changes. | R6 Chief Legal Counsel | Iteration sustains alignment between AI capability and evolving strategy. |